MOHAMMED REHAN SHAIKH

Computer Science Engineering Student | Software Engineering, Backend & Full-Stack
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PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on software development experience through independent projects and hackathons. Builds backend services and REST APIs in Python and FastAPI, backed by PostgreSQL, MongoDB, and MySQL, and deploys them with Docker and Render. Has applied OCR, computer vision, and ML models inside working applications. Seeking Software Engineering, Backend, and Full-Stack roles.

TECHNICAL SKILLS

Languages: Python, Java, SQL
Backend & APIs: FastAPI, REST APIs, JWT Authentication
Databases: PostgreSQL, MongoDB, MySQL
Tools & Platforms: Git, GitHub, Docker, Linux, VS Code, Render
Machine Learning: XGBoost, OCR Pipelines, Gemini API

PROJECTS

CrediMap – Portfolio Builder | Python, FastAPI, MongoDB, Gemini API, JWT

- Built a portfolio platform on FastAPI and MongoDB where users create and publish portfolios from customizable templates.
- Engineered an OCR pipeline to extract text from uploaded certificates, then integrated the Gemini API to identify courses and auto-populate certificate details, reducing manual data entry.
- Implemented JWT-based authentication and public/private portfolio sharing through unique links; deployed to production on Render.

Credit Card Fraud Detection System | Python, FastAPI, PostgreSQL, XGBoost, Docker

- Developed an end-to-end fraud detection application that assesses real-time transaction risk using FastAPI, PostgreSQL, and an XGBoost model.
- Designed a risk-scoring workflow that processes transaction data and flags potentially fraudulent activity.
- Containerized the application with Docker for consistent, reproducible deployment across environments.

HACKATHONS & ACHIEVEMENTS

1st Place – Atharva Hackathon, Dayananda Sagar College of Engineering
Adaptive Cache Management Engine | Python, Random Forest, React.js

- Built an engine that uses a Random Forest model to predict the best cache replacement policy (LRU, LFU, or FIFO) from access frequency, reuse distance, and temporal and spatial locality.
- Delivered a React dashboard showing cache hit rate and latency, with simulated performance gains and exportable PDF reports.

2nd Prize – TechFest Parichay 4.0 (2026), Dept. of AI & ML, T John Institute of Technology
SmartFlow AI: Adaptive Traffic Signal System | Python, FastAPI, YOLOv8, OpenCV

- Replaced fixed-timer signals with real-time control by detecting vehicles from live video and computing lane-wise traffic density.
- Implemented a decision engine that sets green-signal timing (15–60 s), served through FastAPI with a dashboard comparing fixed and AI-controlled signals.

Special Domain Category Prize – ETHiCARE Codeathon (Mar 2026), Dayananda Sagar College of Engineering
Infant Pain Detection System | Python, FastAPI, TensorFlow, React.js, OpenCV, MediaPipe

- Built a system that classifies infant facial expressions as Pain / No Pain with a confidence score, and stores pain history to generate structured medical reports.
- Combined OpenCV and MediaPipe face detection with a MobileNetV2 + LSTM model, integrated through a FastAPI backend, React.js frontend, and SQL database.

HackerRank – 5-Star Rating in Problem Solving

EDUCATION

B.E. Computer Science & Engineering – Dayananda Sagar College of Engineering | CGPA: 8.72 / 10

CERTIFICATION

NPTEL – Computer Networks (Elite), SWAYAM, 2026